



ADNAN KHAN

Operations Research | Stochastic Optimization | Supply Chain Analytics

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📅 October 2026 (Availability) 📍 Paris, France  [GitHub \(click me\)](#)

🎓 EDUCATION

Operations Research & Risk Analytics (M2), CentraleSupélec, Université Paris-Saclay 09/2025 – Present | Paris, France

- Grade: 14/20 (Current Average) | Institutional Scholarship Holder

Industrial and Production Engineering, (B.Eng), University of Dhaka 04/2020 – 12/2024 | Dhaka, Bangladesh

- CGPA: 3.50/4.00 | 87th percentile in the department.

🧠 TECHNICAL SKILLS

Solvers and Modelling — Gurobi, Pyomo, HiGHS, and Excel Solver

Programming and Data — C/C++, Python (NumPy, Pandas, SciPy, Matplotlib), SQL, Power BI

Optimization — MILP, Duality, Sensitivity Analysis, SAA, Benders Decomposition, Vehicle Routing, Network Design

📖 THESIS

Integrated Vehicle Routing, Order Assignment and Optimization for Resilient Omnichannel Distribution Networks,

CentraleSupélec | Supervisor: Prof. Jakob Puchinger | MILP, Vehicle Routing, Python (Pyomo, Gurobi)

- Two-stage stochastic MILP pre-committing backup fulfillment nodes across 3 channels, with routing, 3PL and unserved recourse under 7 disruption scenarios; 14,334 variables, solved in Gurobi to a 1.9% gap.
- Backup layer cut expected cost 32.3% and unserved demand from 5 to 2 units; stochastic planning cut expected cost 61% against the deterministic plan.

📁 EXPERIENCE

Institut du Cerveau (Paris Brain Institute), Intern, Modelling and Data Analysis 06/2026 – Present | Paris, France

Statistical modelling of experimental time series in Python; Gaussian HMMs fitted by maximum likelihood with Viterbi decoding to extract latent states, transition probabilities and dwell times across 32 participants. Built the cleaning and merging pipeline feeding it.

Akij Food and Beverage Limited (AFBL), Industrial Engineering Intern 05/2024 – 07/2024 | Bangladesh

Time and motion studies across 4 layouts to locate throughput bottlenecks; redesigned production layout using LEAN.

Kabir Steel Re-Rolling Mills (KSRM), Production Intern 06/2023 – 08/2023 | Chattogram, Bangladesh

Cut production defects 11% over a 27-day tracked cycle; Pareto and 5-Why root cause analysis to isolate dominant failures.

📁 PROJECTS

Optimization of Maintenance Planning for Offshore Wind Turbines 11/2025 – 12/2025

Python (Pyomo, HiGHS) | MILP, Maintenance Scheduling, Reliability

- MILP scheduling preventive and corrective maintenance across offshore wind farms over a 30 week horizon under crew capacity and travel lockout constraints, solved to optimality in Pyomo with HiGHS; diagnosed from the cost structure that 75€ repair plus transport against a 2€ per week downtime penalty sets the break even at 37.5 weeks, past the horizon.

Capacity Planning and Machine Replacement under Uncertainty 03/2026 – 04/2026

Stochastic Programming, Benders Decomposition, CVaR, Markov Decision Processes

- Capacity planning: two-stage stochastic LP with recourse, solved as an SAA extensive form and by Benders L-shaped decomposition with the two cross-checked against each other; extended to a risk-averse CVaR objective.
- Machine replacement: infinite-horizon MDP over ten deterioration states, value iteration converging in 533 sweeps to a control limit policy that keeps in states 1 to 7 and replaces from state 8; dual LP over discounted occupancy measures priced reliability at about 0.65 k€ per 1% output increase.

Food Bank Inventory Management under Stochastic Demand 01/2026 – 02/2026

Two-Stage SAA, Network Optimization, Equity Constraints, Python

- Extensive-form SAA over 260 demand weeks setting weekly orders for 2 warehouses and the daily 7x2x6 shipment tensor; sparse LP solved with HiGHS. 93.8% fill rate at a binding 10% cap on inter-district satisfaction spread.
- Quantile forecast policy tuned by leave-one-year-out CV: 99.2% fill rate, 1.7% equity gap, no out of sample violations.

🏆 COMPETITIONS

- TOP 10 teams in the BIHRM Supply Chain Contest, achieved 75% merit-based scholarship in SC professional program.
- Regionally selected and competed at the national level in the Bangladesh Math Olympiad (2016).

🌐 LANGUAGES

English - C1 Level (British Council)

French - B1 Level (FLE)

Bengali - Native Level